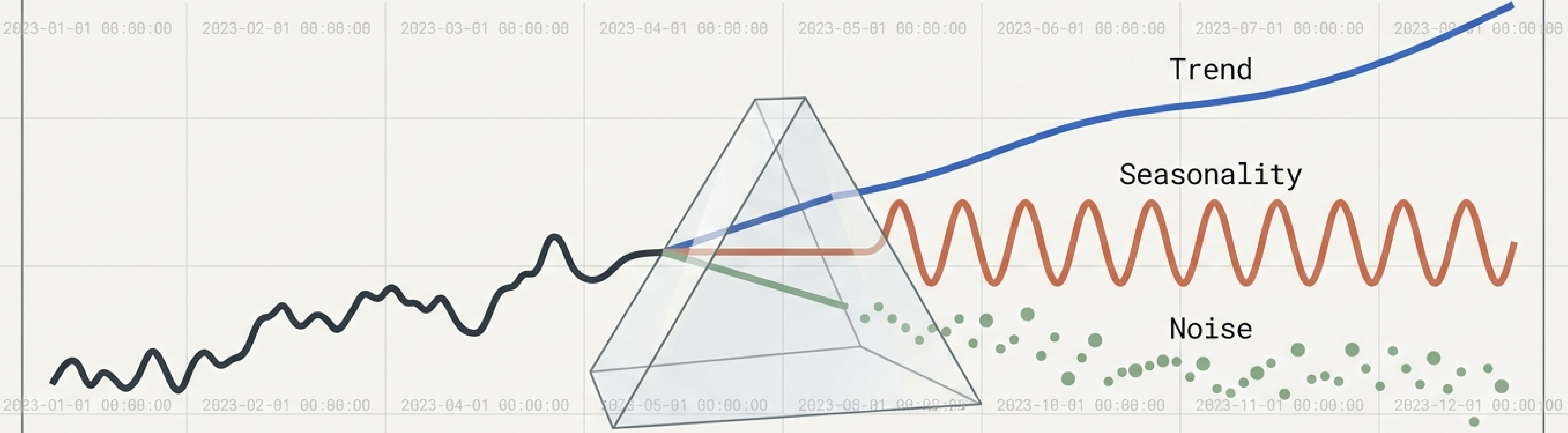
Applied Time Series Analysis: Part II

The Advanced Practitioner's Toolkit: From Transformations to Forecasting and Residual Diagnostics.



The Time Series Analytics Pipeline

1. Data Engineering

- Pandas Time Structures
- Handling Missing Sensor Data
- Computing Log Returns

2. Diagnostics

- Testing for Stationarity
- The ADF Test Flowchart
- Isolating White Noise

3. Forecasting Engines

- Exponential Smoothing
- Meta's Prophet Architecture
- Model Selection Matrix

4. Evaluation

- Validating the Fit
- Residual Diagnostics
- Homoscedasticity Checks

Mastering Time: The Pandas Ecosystem

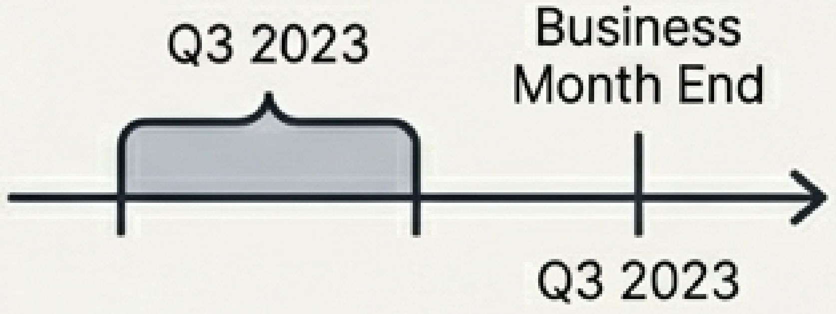
Timestamps

`DatetimeIndex`

Specific moments in time (e.g., July 4th, 2015 at 7:00am). Built on highly efficient 64-bit integers.

Periods

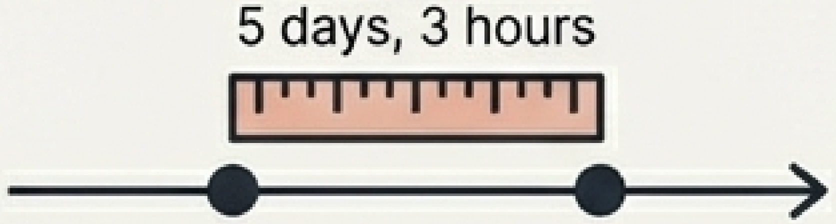
`PeriodIndex`



Fixed-frequency intervals (e.g., Q3 2023, Business Month End).

Deltas

`TimedeltaIndex`



Exact durations and the specific length of time between points.

The Operational Split: Aggregation vs. Selection

resample() = Data Aggregation (e.g., reporting the average value over the previous year).

asfreq() = Data Selection (e.g., reporting the specific exact value at the end of the year).

Mending the Timeline: Handling Missing Data

Forward & Backward Fill

`ffill() / bfill()`

Carries previous or subsequent observations forward/backward. Best for stable states and high-frequency data where the last known state is the most probable.

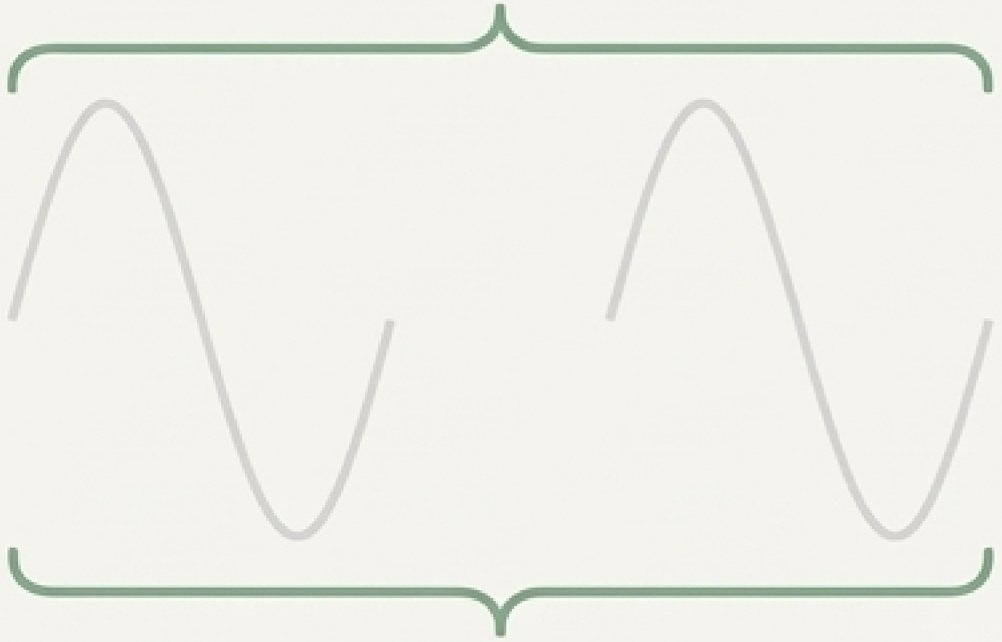
Interpolation

`interpolate()`

Draws a mathematical line between known points. Best for continuous physical phenomena (like temperature) with clear linear trends.

Algorithmic Ignore

`NaN`



Leaving gaps blank. Modern models like Meta's Prophet treat time as a continuous function and natively ignore missing target observations without requiring pre-imputation.

The Case for Log Returns

$$r_t = \ln(p_t/p_{t-1})$$

1. Log-Normality

Bridges raw prices to Gaussian assumptions.

If prices are log-normally distributed, log returns are conveniently normally distributed, allowing classical statistics to apply.

2. Time-Additivity

Radically reduces algorithmic complexity.

Compounding returns over n periods simplifies from O(n) multiplications to O(1) additions.

3. Numerical Stability

Addition of small numbers is numerically safe. Multiplying very small raw returns risks arithmetic underflow in computational systems.

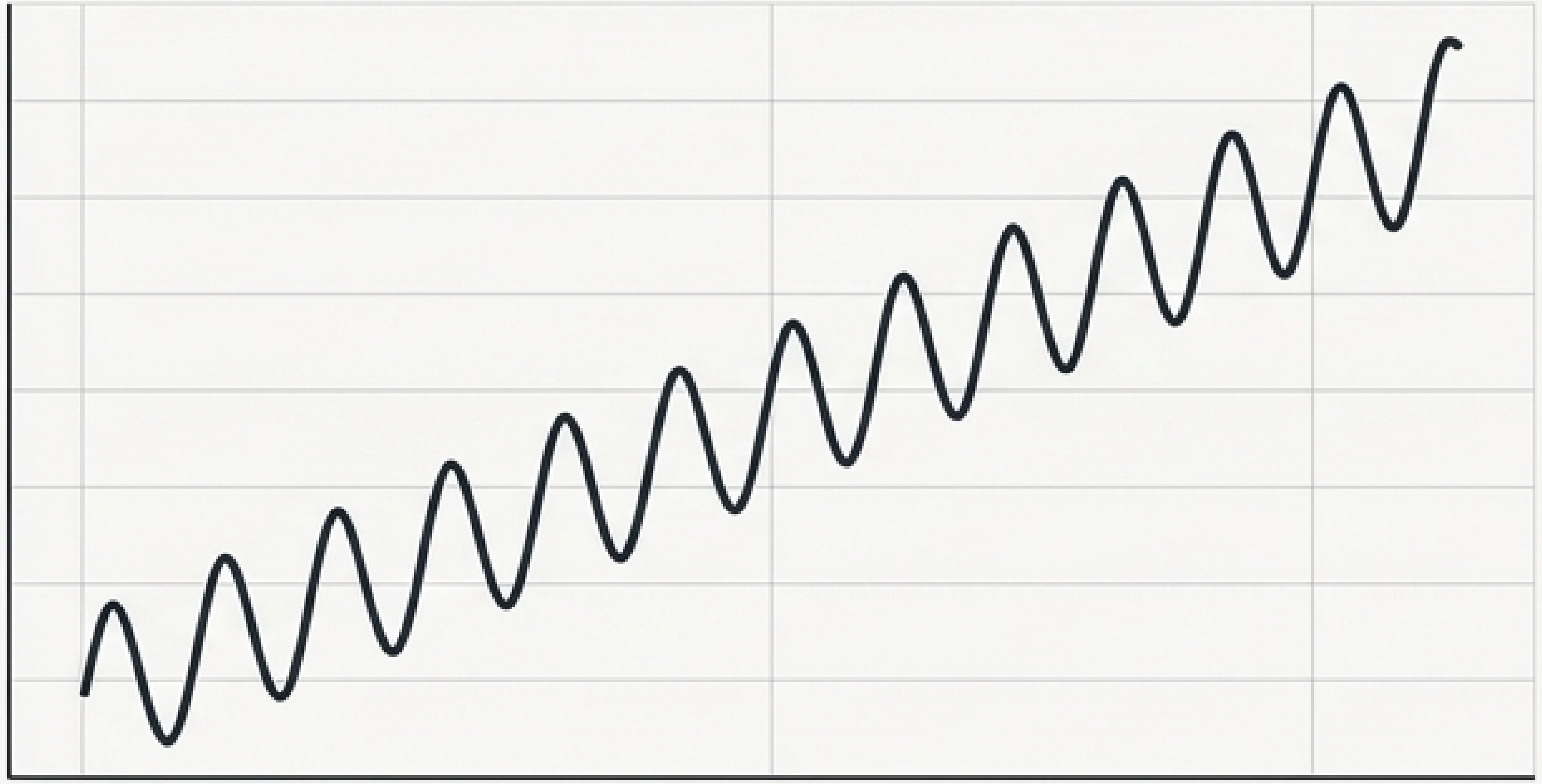
4. Calculus-Friendly

Aligns perfectly with continuous-time stochastic processes, making integration and differentiation mathematically seamless.

Deconstructing the Signal: Time Series Decomposition

Additive Model

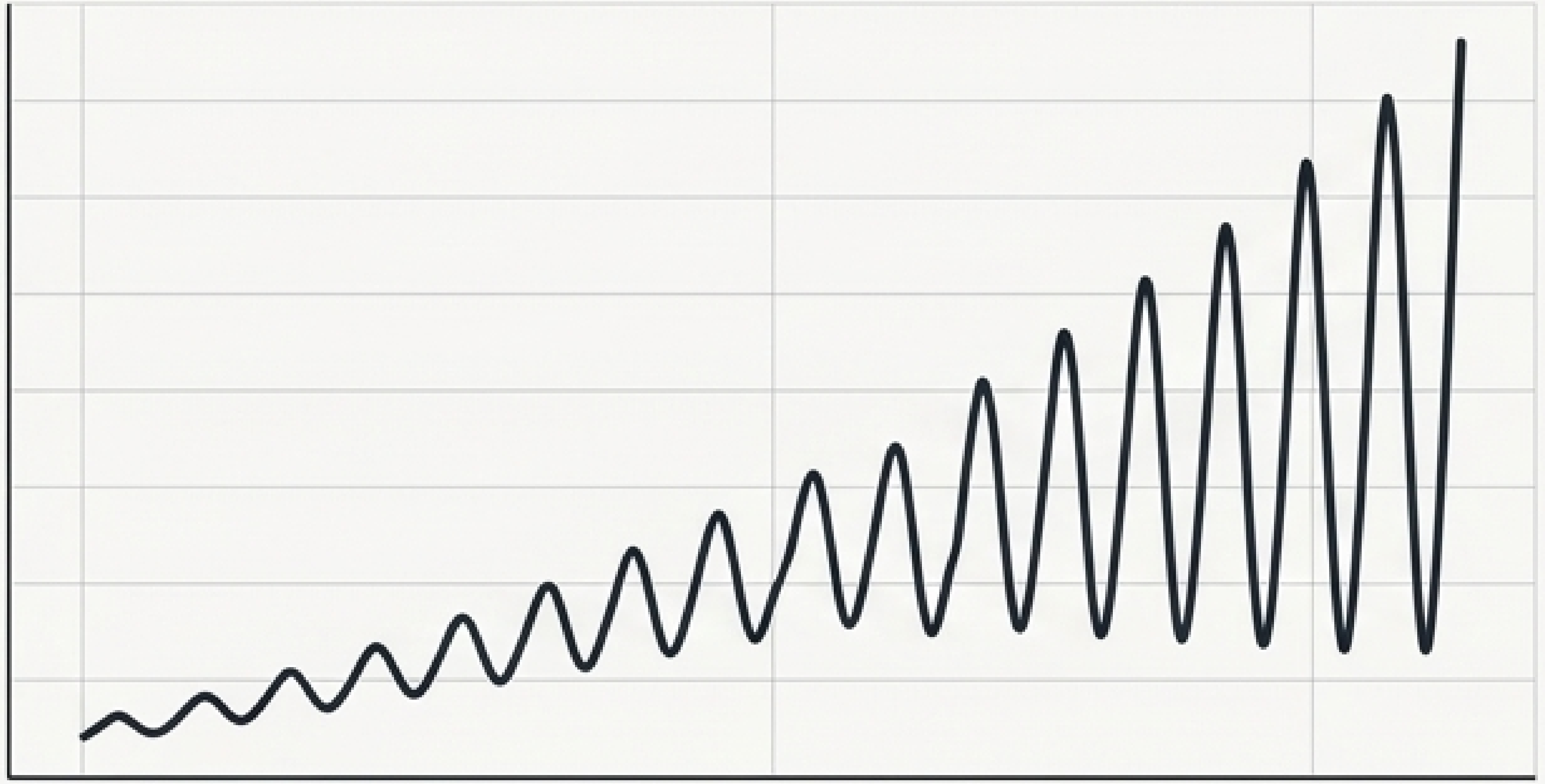
y_t = T_t + S_t + R_t in Roboto Mono



The amplitude of seasonal fluctuations remains constant over time, independent of the overall trend level.

Multiplicative Model

y_t = T_t * S_t * R_t in Roboto Mono



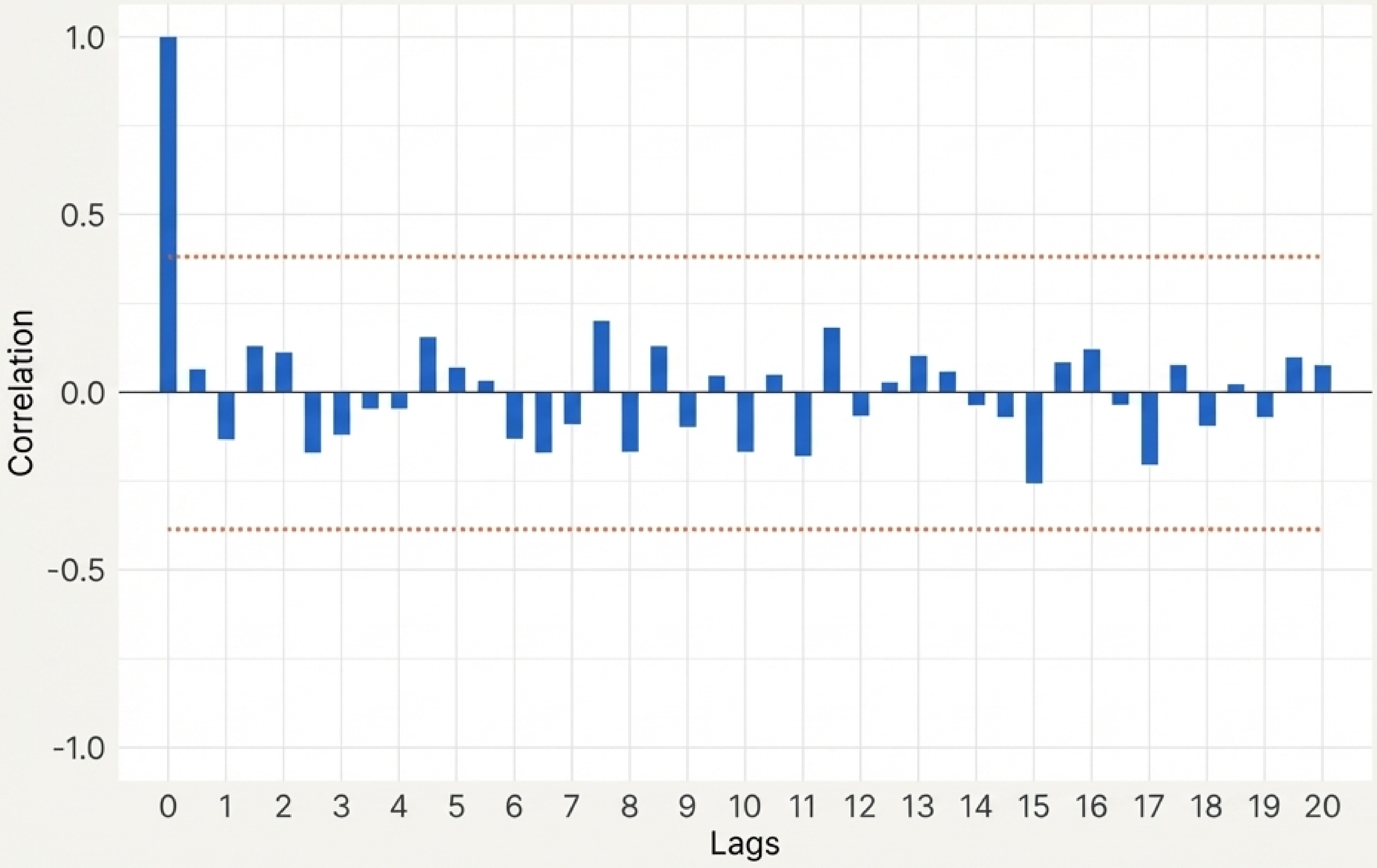
The amplitude of seasonality scales proportionally with the trend (e.g., US Air Passengers or Savings data).

The Transformation Bridge

A log transformation converts a multiplicative process into an additive one, stabilizing the variance:

$\ln(X_t) = \ln(T_t) + \ln(S_t) + \ln(R_t)$ in Roboto Mono

Diagnostic I: Isolating White Noise



Time series that show zero autocorrelation are pure White Noise.

The 95% Rule

For a white noise series, 95% of the spikes in the ACF must lie within the bounds of ± 2/√(T) (where T is the length of the time series).

Example: If `T = 50`, `bounds = ± 0.28`.

If spikes break outside this barrier, the series contains extractble signal.

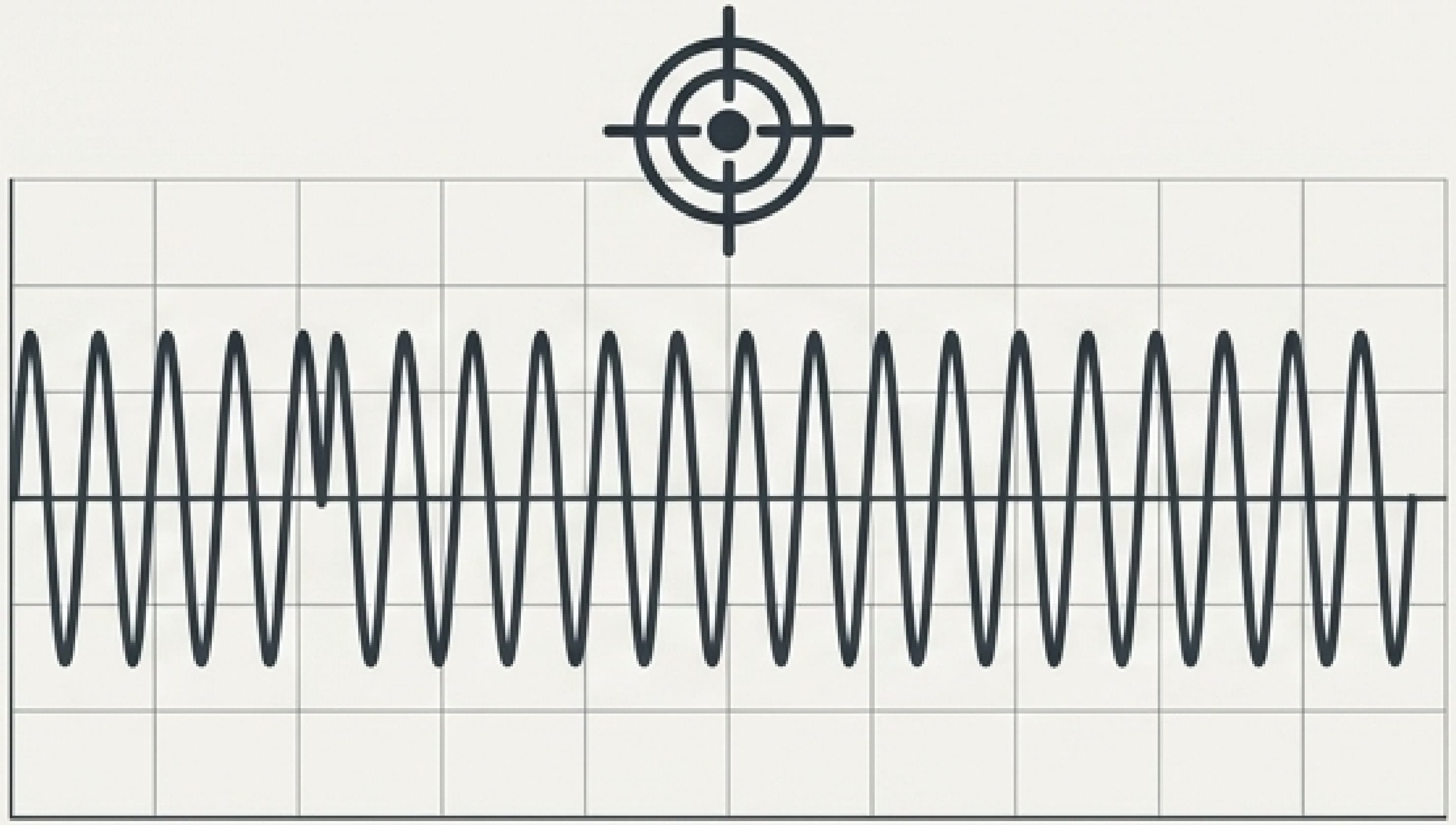
The Premium Data Blueprint

Diagnostic II: The Stationarity Imperative

Non-Stationary (The Problem)

You cannot hit a moving target. A non-stationary series has a wandering mean and expanding variance, leading models to capture spurious correlations.

Stationary (The Requirement)



Weak (Second-Order) Stationarity demands three anchored properties:

1. Constant Mean ($\mu_t = \mu_{t+\tau}$)
2. Finite Variance ($\mathrm{Var}(X_t) < \infty$)
3. Autocovariance depends only on the distance in time between variables, not the time itself.

The Augmented Dickey-Fuller (ADF) Test

$$y_t = \rho\, y_{t-1} + \varepsilon_t$$

Null Hypothesis (H_0): γ = 0

A Unit Root is present.

Result: Non-Stationary

Alt Hypothesis (H_A): γ < 0

No Unit Root is present.

Result: Stationary (Safe to model)

```
statsmodels.tsa.stattools.adfuller
Rule: If p-value < 0.05, REJECT H_0.
Pro-Tip: Set autolag='AIC' to let the algorithm iteratively compute the optimal number
of lags to prevent over-parameterization.
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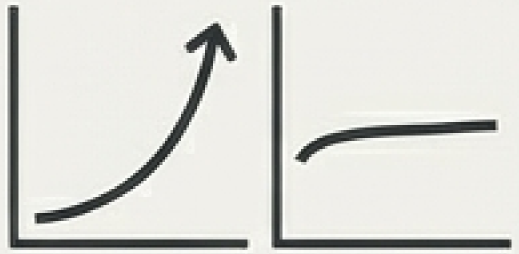
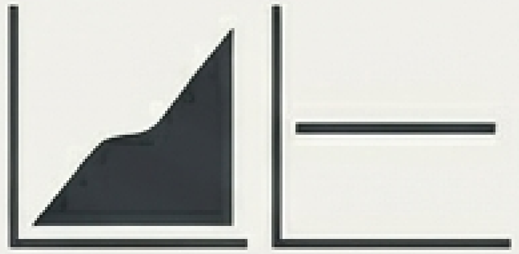
Forging Stationarity: The Transformation Path



Step 1: Stabilize Variance

Apply a Logarithm.

Subdues exponential growth and aligns wildly swinging amplitudes to a consistent scale.



Step 2: Remove Trend

$$\nabla X_t = X_t - X_{t-1}$$

Apply First-Order Differencing.

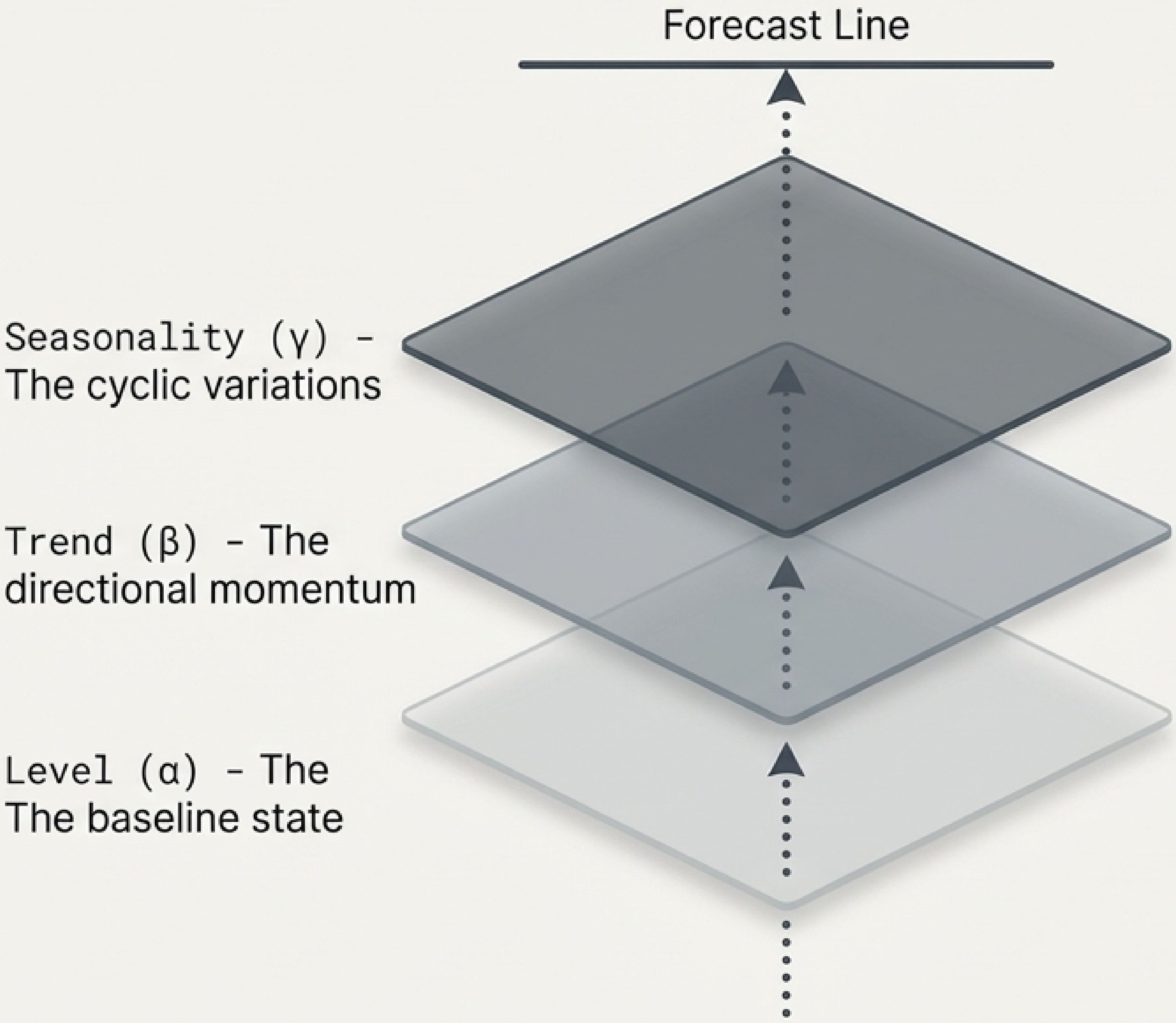
Converts absolute values into period-over-period changes, flattening wandering means to zero.

Step 3: Validate

Re-run ADF test & check ACF/PACF plots.

If the signal is stationary, the pipeline is complete and the data is ready for modeling.

Forecasting Engines I: Exponential Smoothing



Key Insights

The Philosophy:

Curve fitting is (almost) all you need.

The Advantage:

Exceptionally fast and effective for rapid, non-trivial forecasting when data is scarce.

The Limitation:

It is NOT a probabilistic model. It does not natively calculate confidence intervals or posterior distributions, assuming past patterns will strictly continue.

Forecasting Engines II: Meta’s Prophet

The Premium Data Blueprint

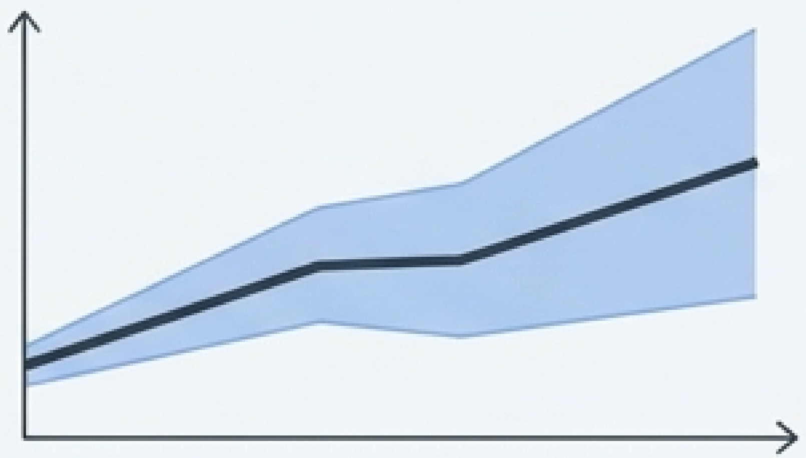
Data Requirements

Requires strict naming: `"ds" (datastamps)` and `"y" (target)`.

Core Capabilities

- Natively ignores gaps/missing data in the target variable.
- Ingests external regressors (covariates).
- Accommodates overlapping multiple seasonalities (weekly + yearly) using Fourier series.

Probabilistic Forecast



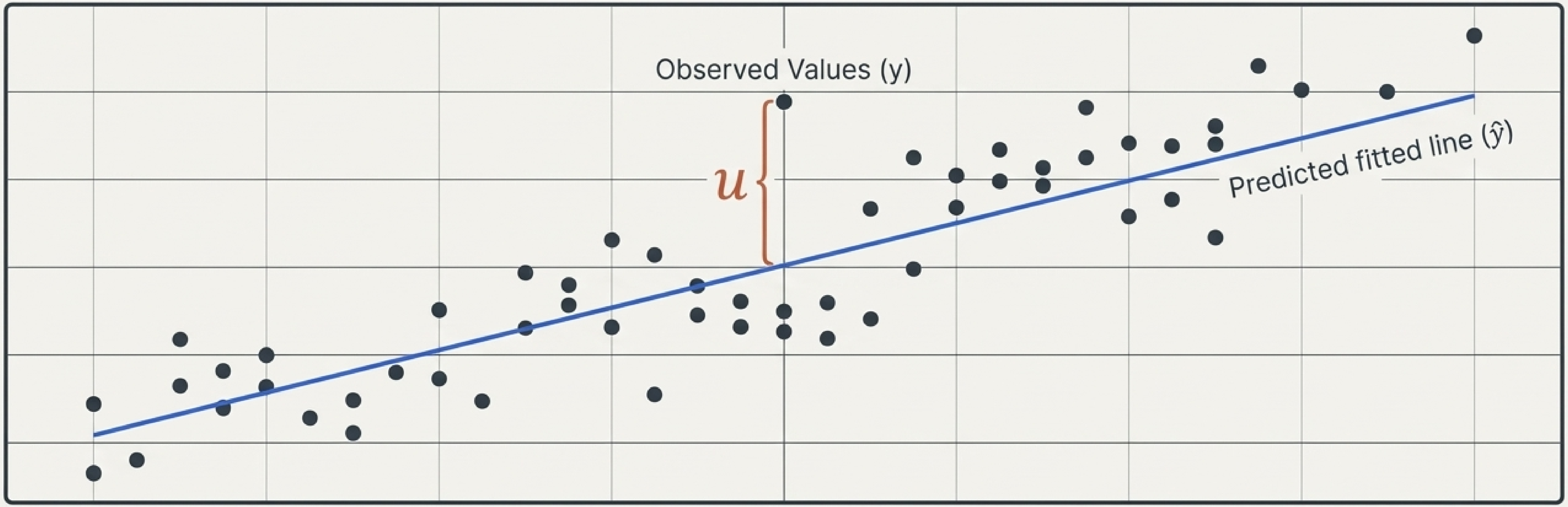
Generates point forecasts wrapped in MCMC (Monte Carlo Markov Chain) simulated confidence bands, providing explicit uncertainty measurement.

Forecasting Engine Comparison Matrix

Criteria	ARIMA	Exponential Smoothing	Meta's Prophet
Stationarity Required?	Yes (Strictly)	No	No
Handles Missing Data?	Requires Imputation	Requires Imputation	Natively Ignores Gaps
Probabilistic Output?	Yes (Statistical CI)	No (Point forecasts only)	Yes (MCMC generated bounds)
Speed / Computation	Slower (Iterative optimization)	Instantaneous	Fast (Slows with MCMC sampling)

Evaluation: The Power of Residuals

The Premium Data Blueprint

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Residual (u) = Observed Value (y) - Predicted Value (ŷ)
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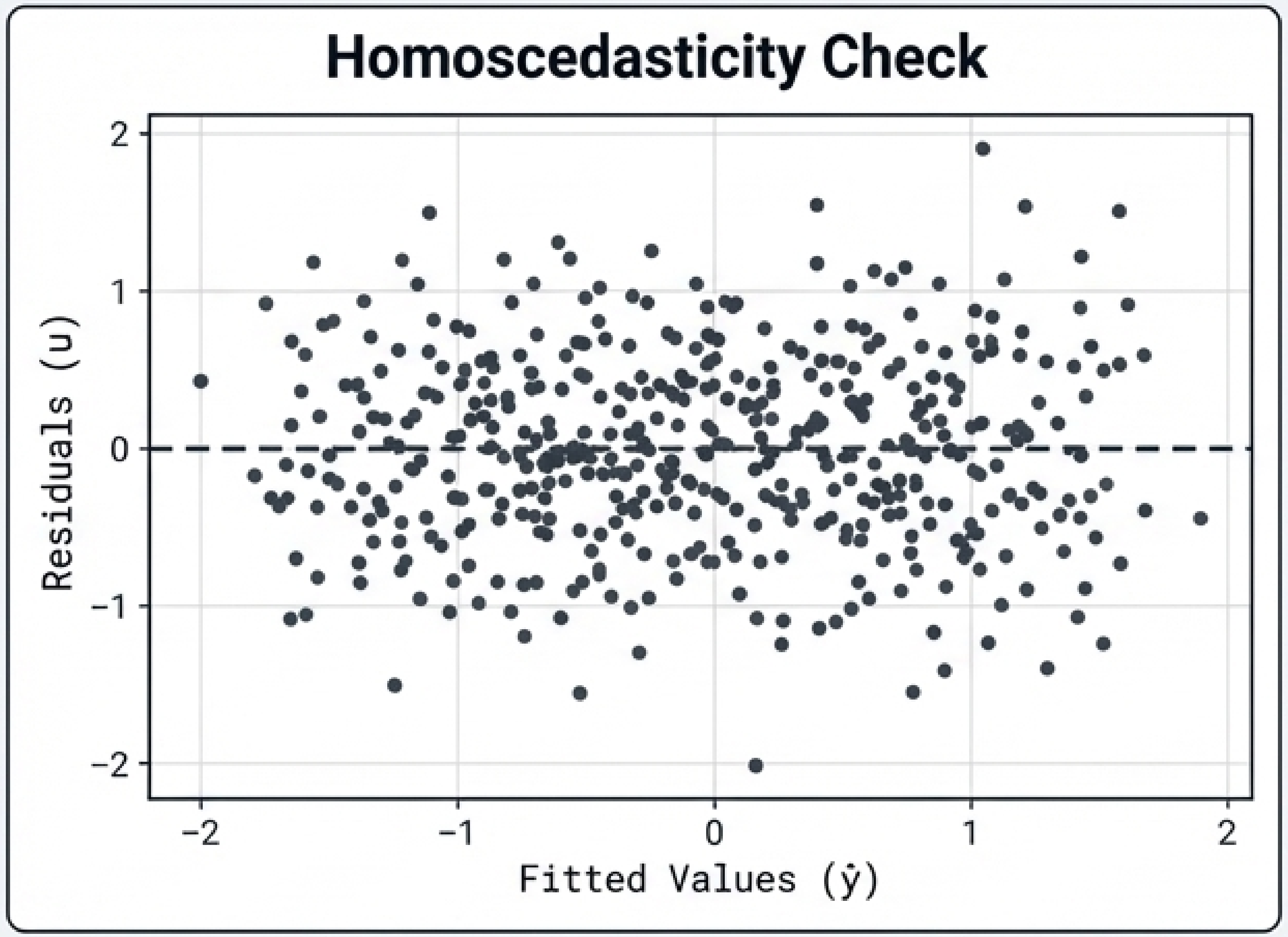
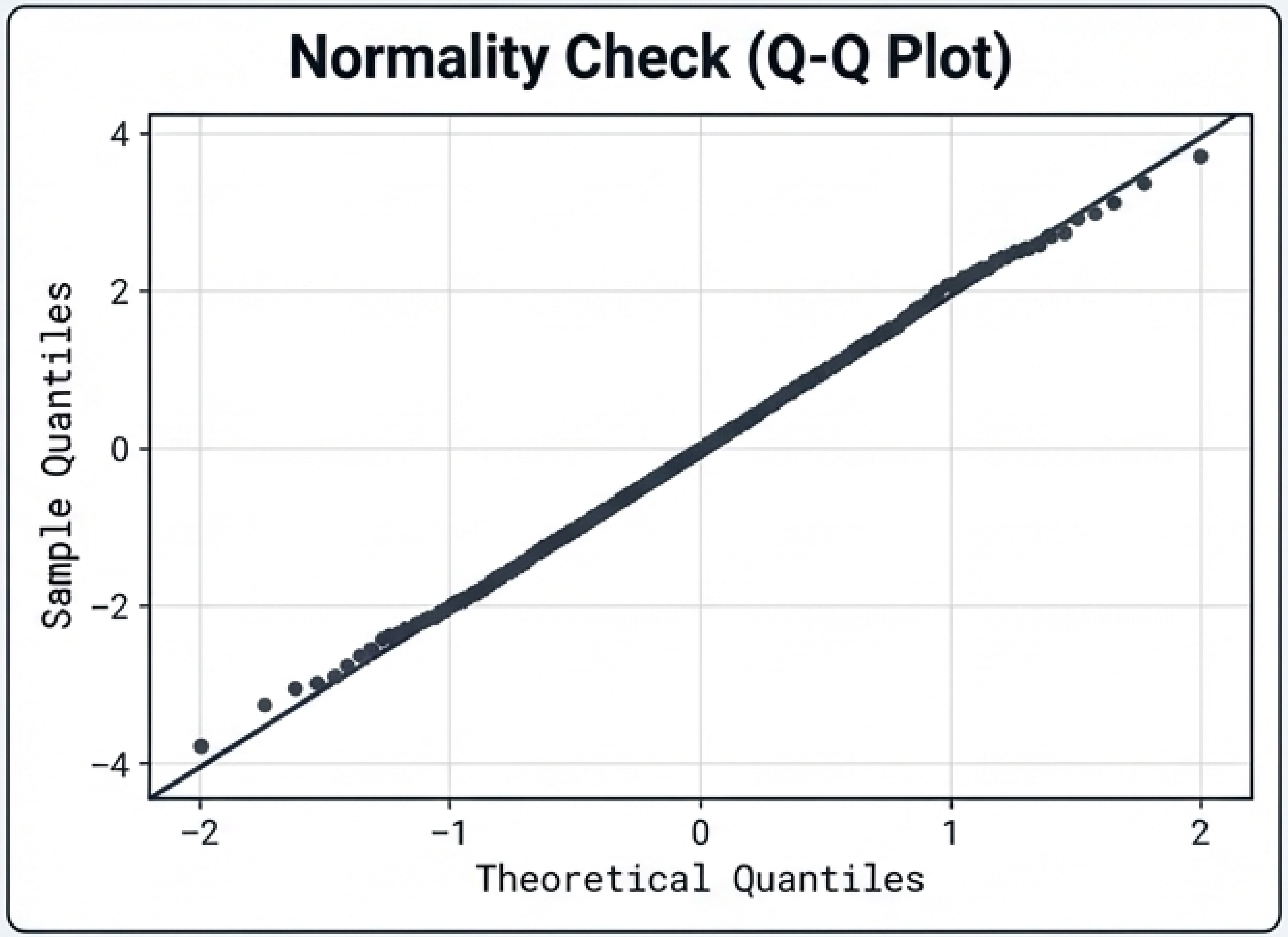
	Positive Residual: Observation is above the prediction line.
	Negative Residual: Observation is below the prediction line.
	Golden Rule: For a properly fitted linear model, the sum of all residuals equals zero, and their mean is exactly zero.

Residual Diagnostics: Validating the Fit



Look for a random, even scatter around the zero-line. **Warning:** A ‘cone’ shape indicates Heteroscedasticity (changing variance), meaning the model is unstable.



If the plotted points form a straight diagonal line, the assumption of normal distribution is successfully met.

If your residuals aren’t purely White Noise, your model has left valuable signal on the table.